

I Theory Appendix: Proofs and Details

The sufficient statistic rests on an envelope theorem for the consumer: when interchange fees change, consumers care only about the change in retail prices on the goods they buy and the change in the reward on the method they use. The hard part of the proof is to show that those equilibrium price changes are well approximated by functions of equilibrium quantities we can measure at the merchant—the composition of payments and the fee schedule.

I.A Setup and Notation

We prove Theorem ?? in the sufficient-statistic environment of Section ?. Consumers, merchants, and networks behave as in that section: type- k consumers maximize U_k subject to $\sum_j p_j q_{jk} \leq 1 + f_k$; interchange fees follow (?); retail prices follow (?); and rewards follow (?). We use stars for equilibrium values and index payment methods by k and l .

For the proofs it is convenient to write each merchant's payment-weighted average fee as $\hat{\tau}_j \equiv \sum_m \mu_{jm} \tau_{jm}$, so (?) reads $\log p_j = \log \bar{p}_j + \rho^P \hat{\tau}_j$. Revenue weights ω_j and the operator $\mathbb{E}_R[\cdot] \equiv \sum_j \omega_j(\cdot)$ are as in Theorem ?; let $\tau_{\max} \equiv \max_{j,k} |\tau_{jk}|$. Rewards inflate each cohort's effective budget to $1 + f_k$, so aggregate expenditure is $1 + O(\tau_{\max})$ (Assumption 2); in all first-order expressions below we treat aggregate expenditure as one and absorb the $O(\tau_{\max})$ correction into the stated remainders. The proof uses the following additional notation.

Symbol	Dimension	Meaning
$\hat{\tau}_j$	scalar	merchant j 's payment-weighted average fee, $\sum_m \mu_{jm} \tau_{jm}$
$\tau_{jk} - \hat{\tau}_j$	scalar	price-side “excess fee”: method k 's deviation from merchant j 's average
s_{jk}^*	scalar	cohort k 's spending share at merchant j , $p_j q_{jk}^* / \sum_{j'} p_{j'} q_{j'k}^*$
$\bar{\tau}^k$	scalar	cohort k 's spending-weighted average fee, $\sum_j s_{jk}^* \tau_{jk}$
E^k	$\mathbb{R}^{J \times J}$	price elasticities of cohort k : $E_{ji}^k = \partial \log q_j^k / \partial \log p_i$
G^k	$\mathbb{R}^{J \times K}$	income semi-elasticities of cohort k : $G_{jm}^k = \partial \log q_j^k / \partial f_m$

I.B Assumptions

Let $E_{ji}^k \equiv \partial \log q_j^k / \partial \log p_i$ denote the cross-price elasticity of demand among consumers of type k , and let E stack these matrices across types.

Assumption 1. *The elasticities E_{ji}^k are finite and satisfy a dominant diagonal condition: there exist $M, m > 0$ such that for each k and j ,*

$$|E_{jj}^k| < M \quad \text{and} \quad \sum_{i \neq j} |E_{ji}^k| < m |E_{jj}^k|.$$

Own-price elasticities are bounded above and dominate the sum of cross-price elasticities for the same good. CES demand with $\sigma > 1$ satisfies the condition with $M = \sigma$ and $m = 1$.

Assumption 2 (Linear Reward Pass-Through). *Rewards follow (??) with a constant marginal pass-through ρ^R and a policy-invariant baseline, $d\bar{f}_k/dx_l = 0$ for every fee parameter x_l . Observed rewards are the same order of magnitude as the fees, $f_k = O(\tau_{\max})$.*

The baseline $\bar{f}_k$ collects network costs, issuer margins, and any cross-subsidies that do not move with the fee schedule; because it is differenced out of every policy derivative, its level never enters the welfare statistic. The only technical content the proof draws from this assumption is that observed rewards are the same order as fees—a directly measurable property (rewards run about 1–2% of spend in the data, the same order as interchange) rather than a bound on unobserved costs.

Assumption 3 (Bounded Income Semi-Elasticities). *The demand of a consumer of type k admits finite income semi-elasticities $G_{jm}^k \equiv \partial \log q_j^k / \partial f_m$, with row sums bounded uniformly: $\sum_m |G_{jm}^k| \leq L$ for some finite L and for all j and k .*

Homothetic demand is a sufficient special case, with $G_{jm}^k = \mathbb{1}\{m = k\} / (1 + f_k)$ and hence $L = 1$. The three demand bounds M , m , and L are the only primitives that enter the theorem: M controls how demand responds to retail prices, m controls how demand responds to cross-price elasticities, and L controls how demand responds to rewards (an income change for the cohort). All three reappear in the contraction threshold of Step C.

I.C Proof of Theorem ??

The proof has four steps. Step A applies the envelope theorem to reduce welfare to a price term and a reward term. Step B evaluates the reward term from (??). Step C gives an explicit bound showing that equilibrium feedback is small relative to the direct price and reward effects. Step D assembles the pieces.

Step A: Envelope theorem. Roy’s identity applied to the consumption problem of a consumer in cohort k gives

$$\frac{1}{\lambda_k} \frac{dV_k}{dx_l} = - \sum_j p_j q_{jk}^* \frac{d \log p_j}{dx_l} + \frac{df_k}{dx_l}.$$

Because the consumer was already optimizing, reallocating her basket has no first-order welfare effect; she is hit only through retail prices on the goods she buys, weighted by her spending, and through the reward on the method she uses.

Since aggregate expenditure is $1 + O(\tau_{\max})$, the spending weight $p_j q_{jk}^*$ equals cohort k ’s spending share s_{jk}^* up to $O(\tau_{\max})$, so Roy’s identity reads

$$\frac{1}{\lambda_k} \frac{dV_k}{dx_l} = - \sum_j s_{jk}^* \frac{d \log p_j}{dx_l} + \frac{df_k}{dx_l} + O(\tau_{\max}). \quad (1)$$

We carry out the proof in these cohort spending shares—the natural weights for the envelope and reward steps—and translate the result into the observable revenue-weighted moments of Theorem ?? only at the end (Step D). The bridge is a spending identity: cohort k spends $\mu_k p_j q_{jk}^*$ at merchant j , while merchant j draws the share $\omega_j \mu_{jk}$ of economy-wide revenue from method k ; equating the two accounting conventions gives

$$\mu_k s_{jk}^* = \omega_j \mu_{jk} + O(\tau_{\max}), \quad (2)$$

where the remainder collects reward-induced deviations from unit aggregate expenditure. Equivalently, a cohort spending-weighted sum is a revenue-weighted moment, $\sum_j s_{jk}^*(\cdot) = \mu_k^{-1} \mathbb{E}_R[\mu_{jk}(\cdot)] + O(\tau_{\max})$. It remains to characterize the reward response df_k/dx_l (Step B) and the price response $d \log p_j/dx_l$ (Step C).

Step B: Reward response (level identity). The reward rule (??) is itself a level identity in spending shares, $f_k = \bar{f}_k + \rho^R \bar{\tau}^k$ with $\bar{\tau}^k = \sum_j s_{jk}^* \tau_{jk}$: rewards on method k equal the fixed baseline $\bar{f}_k$ plus the marginal pass-through ρ^R times the average fee that cohort k 's spending encounters.

Differentiate in x_l . The baseline is policy-invariant, $d\bar{f}_k/dx_l = 0$ (Assumption 2), and $d\tau_{jk}/dx_l = \mathbb{1}\{k = l\} \partial \tau_{jl} / \partial x_l$, so

$$\frac{df_k}{dx_l} = \rho^R \left(\mathbb{1}\{k = l\} \sum_j s_{jk}^* \frac{\partial \tau_{jl}}{\partial x_l} + \sum_j \tau_{jk} \frac{ds_{jk}^*}{dx_l} \right).$$

The second sum pairs the fee $\tau_{jk} = O(\tau_{\max})$ with the movement of the spending shares; recentering it against $\bar{\tau}^k$ (the shares sum to one) and using the explicit feedback bound from Step C, it is $O(\tau_{\max})$. Hence

$$\frac{df_k}{dx_l} = \rho^R \mathbb{1}\{k = l\} \sum_j s_{jl}^* \frac{\partial \tau_{jl}}{\partial x_l} + O(\tau_{\max}). \quad (3)$$

Equation (3) is the network-balance identity: every dollar the network collects from the rate hike on method l is rebated to method- l users, and other cohorts are unaffected at first order. Because the baseline $\bar{f}_k$ is fee-invariant, it drops out of the derivative and leaves the reward response—and hence Theorem ??—unchanged; the level of network costs and issuer margins never enters.

Step C: Equilibrium feedback (explicit bound). Differentiating (??)—equivalently, $\log p_j = \log \bar{p}_j + \rho^P \hat{\tau}_j$ —and recentering the composition shift with $\sum_m d\mu_{jm}/dx_l = 0$ (shares sum to one)

gives

$$\frac{d \log p_j}{dx_l} = \rho^P \left[\underbrace{\mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l}}_{\text{direct effect}} + \underbrace{\sum_m (\tau_{jm} - \hat{\tau}_j) \frac{d \mu_{jm}}{dx_l}}_{\text{composition feedback}} \right]. \quad (4)$$

The direct term is the mechanical pass-through of the fee into prices. The composition feedback is small because it multiplies share movements by centered fee gaps, each of which is at most order $\tau_{\max}$.

Let

$$z = \begin{pmatrix} d \log p \\ df \end{pmatrix}, \quad z_0 = \begin{pmatrix} z_0^p \\ z_0^f \end{pmatrix},$$

where

$$z_{0,j}^p = \rho^P \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l}, \quad z_{0,k}^f = \rho^R \mathbb{1}\{k = l\} \sum_j s_{jl}^* \frac{\partial \tau_{jl}}{\partial x_l}.$$

Thus z_0 is the response holding payment shares and spending shares fixed. Let

$$B_l \equiv \max_j \left| \frac{\partial \tau_{jl}}{\partial x_l} \right|, \quad H \equiv (1 + m)M + L, \quad C \equiv 2(1 + H). \quad (5)$$

Because $\rho^P, \rho^R \leq 1$, $\mu_{jl} \leq 1$, and $\sum_j s_{jl}^* = 1$, $\|z_0\|_\infty \leq B_l$.

It remains to bound $z - z_0$. Demand responds to prices and rewards according to

$$\frac{d \log q_j^k}{dx_l} = \sum_i E_{ji}^k \frac{d \log p_i}{dx_l} + \sum_n G_{jn}^k \frac{df_n}{dx_l}.$$

The row-sum bounds in Assumptions 1 and 3 imply, for every j and k ,

$$\left| \frac{d \log q_j^k}{dx_l} \right| \leq H \|z\|_\infty. \quad (6)$$

For prices, the within-merchant payment share $\mu_{jm} = \mu_m q_{jm}^* / \sum_n \mu_n q_{jn}^*$ nets out the common price p_j , so the quotient rule gives

$$\frac{d \log \mu_{jm}}{dx_l} = \frac{d \log q_j^m}{dx_l} - \sum_n \mu_{jn} \frac{d \log q_j^n}{dx_l}.$$

Since $\sum_m (\tau_{jm} - \hat{\tau}_j) \mu_{jm} = 0$, the average term drops out of the recentered composition feedback:

$$\sum_m (\tau_{jm} - \hat{\tau}_j) \frac{d \mu_{jm}}{dx_l} = \sum_m (\tau_{jm} - \hat{\tau}_j) \mu_{jm} \frac{d \log q_j^m}{dx_l}.$$

The centered merchant fee gap satisfies

$$\sum_m |(\tau_{jm} - \hat{\tau}_j) \mu_{jm}| \leq 2\tau_{\max}.$$

Combining this inequality with (6),

$$\left| \frac{d \log p_j}{dx_l} - \rho^P \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right| \leq 2H\tau_{\max} \|z\|_{\infty}. \quad (7)$$

For rewards, the feedback comes from the spending-share term left in Step B. The spending share $s_{jk}^* = p_j q_{jk}^* / \sum_{j'} p_{j'} q_{j'k}^*$ has derivative

$$\frac{d \log s_{jk}^*}{dx_l} = \left(\frac{d \log p_j}{dx_l} + \frac{d \log q_j^k}{dx_l} \right) - \sum_{j'} s_{j'k}^* \left(\frac{d \log p_{j'}}{dx_l} + \frac{d \log q_{j'}^k}{dx_l} \right).$$

Recentering by $\bar{\tau}^k$ makes the average term vanish:

$$\sum_j \tau_{jk} \frac{ds_{jk}^*}{dx_l} = \sum_j s_{jk}^* (\tau_{jk} - \bar{\tau}^k) \left(\frac{d \log p_j}{dx_l} + \frac{d \log q_j^k}{dx_l} \right).$$

The centered cohort fee gap satisfies

$$\sum_j |s_{jk}^* (\tau_{jk} - \bar{\tau}^k)| \leq 2\tau_{\max}.$$

Therefore

$$\left| \frac{df_k}{dx_l} - \rho^R \mathbb{1}\{k = l\} \sum_j s_{jl}^* \frac{\partial \tau_{jl}}{\partial x_l} \right| \leq 2(1 + H)\tau_{\max} \|z\|_{\infty}. \quad (8)$$

Equations (7) and (8) together imply

$$\|z - z_0\|_{\infty} \leq C\tau_{\max} \|z\|_{\infty}, \quad C \equiv 2[1 + (1 + m)M + L]. \quad (9)$$

Set $\bar{\tau} \equiv 1/C$. The same calculation applies to the difference between any two candidate responses, so whenever $\tau_{\max} < \bar{\tau}$, the feedback map is a contraction. For the equilibrium response,

$$\|z\|_{\infty} \leq \|z_0\|_{\infty} + \|z - z_0\|_{\infty} \leq B_l + C\tau_{\max} \|z\|_{\infty},$$

so

$$\|z\|_{\infty} \leq \frac{B_l}{1 - C\tau_{\max}} \quad \text{and} \quad \|z - z_0\|_{\infty} \leq \frac{C\tau_{\max}}{1 - C\tau_{\max}} B_l.$$

The coordinate bounds are therefore

$$\left| \frac{d \log p_j}{dx_l} - \rho^P \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right| \leq \frac{2H\tau_{\max}}{1 - C\tau_{\max}} B_l, \quad (10)$$

$$\left| \frac{df_k}{dx_l} - \rho^R \mathbb{1}\{k = l\} \sum_j s_{jl}^* \frac{\partial \tau_{jl}}{\partial x_l} \right| \leq \frac{2(1 + H)\tau_{\max}}{1 - C\tau_{\max}} B_l. \quad (11)$$

For the fee components used below, $B_l = 1$. Hence the feedback is $O(\tau_{\max})$ in both coordinates,

$$\frac{d \log p_j}{dx_l} = \rho^P \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} + O(\tau_{\max}), \quad (12)$$

and the share movements left over in Step B are likewise $O(\tau_{\max})$, confirming (3). This threshold $\bar{\tau} = 1/C$ is the explicit threshold the theorem promises.

Step D: Assembly. Substitute the price response (12) and the reward response (3) into the spending-share envelope formula (1):

$$\frac{1}{\lambda_k} \frac{dV_k}{dx_l} = -\rho^P \sum_j s_{jk}^* \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} + \mathbb{1}\{k = l\} \rho^R \sum_j s_{jk}^* \frac{\partial \tau_{jk}}{\partial x_k} + O(\tau_{\max}).$$

Now translate the cohort spending-weighted sums into observable revenue-weighted moments with the spending identity (2), $\sum_j s_{jk}^*(\cdot) = \mu_k^{-1} \mathbb{E}_R[\mu_{jk}(\cdot)] + O(\tau_{\max})$:

$$\frac{1}{\lambda_k} \frac{dV_k}{dx_l} = \mu_k^{-1} \left(-\rho^P \mathbb{E}_R \left[\mu_{jk} \mu_{jl} \frac{\partial \tau_{jl}}{\partial x_l} \right] + \mathbb{1}\{k = l\} \rho^R \mathbb{E}_R \left[\mu_{jk} \frac{\partial \tau_{jk}}{\partial x_k} \right] \right) + O(\tau_{\max}), \quad (13)$$

which is equation (??) and proves Theorem ??.

□

I.D Proof of Fee Component Formulas

We derive the fee component decomposition by applying Theorem ?? to specific counterfactual changes in the fee structure. We set $\rho^P = \rho^R = 1$ throughout for simplicity in writing the expressions. The welfare effect of any fee change equals the size of the change multiplied by the derivative from Theorem ??, so it is enough to track the derivative for each component.

Baseline Fees. The baseline fee $\bar{\tau}_l$ shifts every interchange rate in (??) by the same amount, so $\partial \tau_{jl} / \partial \bar{\tau}_l = 1$ for all merchants. Applying Theorem ?? and using $\mathbb{E}_R[\mu_{jk}] = \mu_k + O(\tau_{\max})$,

$$A_{kl} \equiv \bar{\tau}_l \frac{1}{\lambda_k} \frac{dV_k}{d\bar{\tau}_l} = \bar{\tau}_l \mu_k^{-1} (-\mathbb{E}_R[\mu_{jk} \mu_{jl}] + \mathbb{1}\{k = l\} \mu_k).$$

Sector Adjustments. Sector adjustments shift fees only for merchants in the relevant sector, so $\partial \tau_{jl} / \partial \tau_{sl} = \mathbb{1}\{s_j = s\}$. Write $\omega_s = \sum_{s_j=s} \omega_j$ for sector s 's share of total revenue and

$\mu_{sl} = \mathbb{E}_R[\mu_{jl} \mid s_j = s]$ for its average share on card l . The normalization $\sum_s \omega_s \mu_{sl} \tau_{sl} = 0$ makes the reward term vanish, because

$$\mathbb{E}_R[\mu_{jl} \tau_{s_j, l}] = \sum_s P_R(s_j = s) \mathbb{E}_R[\mu_{jl} \mid s_j = s] \tau_{sl} = 0.$$

Only the price (second-moment) term remains, and the total effect of sector adjustments is

$$\sum_s \tau_{sl} \frac{1}{\lambda_k} \frac{dV_k}{d\tau_{sl}} = -\mu_k^{-1} \sum_s \tau_{sl} P_R(s_j = s) \mathbb{E}_R[\mu_{jk} \mu_{jl} \mid s_j = s].$$

Negotiated Rates. The negotiated discount δ_l shifts the fee only for large merchants, so $\partial \tau_{jl} / \partial \delta_l = b_j$, which equals one for large merchants and zero otherwise. Applying Theorem ??,

$$B_{kl} \equiv \delta_l \frac{1}{\lambda_k} \frac{dV_k}{d\delta_l} = \delta_l \mu_k^{-1} P_R(b_j = 1) (-\mathbb{E}_R[\mu_{jk} \mu_{jl} \mid b_j = 1] + \mathbb{1}\{k = l\} \mathbb{E}_R[\mu_{jk} \mid b_j = 1]).$$